

Unequal- μ product floors and the $n=5$ polygon floor

Chase Hendrick

Found / derived in the SharpMeow/GENChase validation campaign (2026).

Cite the author by name. Do not attribute to the GitHub handle SharpMeow.

Do not treat this note as a personal theorem stamp (no “Hendrick’s Identity / Law”).

Version 2, revised 2026-09-23 (America/New_York). Companion markdown and the validation script are in-repo under `research/`.

Abstract

We record exact closed forms for classical point-vortex collapse floors, where the floor is the minimum of the spin-collapse product $\omega_0 t_c$ (initial rotation rate times collapse time) over a self-similar collapsing family. (i) For $\Gamma = (1, 1/2, -1/3)$ the angular-impulse-zero ($L = 0$) family has **two** collapsing branches, one for each orientation of the triangle. Their floors are the two positive real roots of one irreducible sextic: $P_{\min} \approx 1.064706$ (the global floor) and $P_\star \approx 2.203855$ (the floor on the branch covered by Gotoda’s $L = 0$ arc, which version 1 of this note reported as the $\mu = 1/2$ floor). (ii) The $n = 5$ specialization $F_5 = \sqrt{31682}/80$ of the two-ring polygon floor F_n . Both results were re-derived from raw Biot–Savart velocities, independently of the formulas they replace, and the algebra was checked in exact arithmetic. The families and the product $\omega_0 t_c$ are classical (Gröbli, Aref, Gotoda, Koiller et al.); only the minimizations and their closed forms are claimed. They are **proved candidates; priority is unconfirmed**.

Correction to version 1. Version 1 called $P_\star \approx 2.2039$ “the $\mu = 1/2$ product floor”. $P_\star$ is the minimum on Gotoda’s arc $0 < \theta < \theta_0$ only. The opposite orientation, $\pi < \theta < 2\pi - \theta_0$ in the same parametrization, also collapses, and its minimum $P_{\min} \approx 1.0647$ is lower. The original derivation found this critical point ($\cos \theta \approx -0.9244$) and set it aside as “on the expanding side”; that is true for $\sin \theta > 0$, but with $\sin \theta < 0$ the configuration collapses. Every algebraic statement about $P_\star$ in version 1 remains correct.

1. Statements

1.1. $\mu = 1/2$: two collapsing branches, two floors

Place the vortices $\Gamma = (1, 1/2, -1/3)$ on Gotoda’s $L = 0$ circle construction with parameter θ and the 2π Biot–Savart kernel. The product is

$$P(\theta) = \omega_0 t_c = \frac{14 \sin^2 \theta + 6\sqrt{7} \cos \theta + 21}{2(14 \cos \theta + \sqrt{7}) \sin \theta}.$$

With $\cos \theta_0 = -\sqrt{7}/14$, the configuration collapses exactly on two open arcs:

- **Branch B** (Gotoda’s arc): $0 < \theta < \theta_0 \approx 100.89^\circ$.
- **Branch A** (opposite orientation): $\pi < \theta < 2\pi - \theta_0 \approx 259.11^\circ$.

On each arc $P > 0$, $P \rightarrow +\infty$ at both ends, and P has exactly one critical point. With

$$X = \frac{245351\sqrt{5201}}{5201^2}, \quad R = \frac{7\sqrt{5201}}{162},$$

the two floors are

$$P_{\min} = \sqrt{\frac{605}{324} + R \cos\left(\frac{1}{3} \arccos X - \frac{2\pi}{3}\right)} \approx 1.064705976271204,$$

$$P_{\star} = \sqrt{\frac{605}{324} + R \cos\left(\frac{1}{3} \arccos X\right)} \approx 2.203855016036133.$$

$P_{\min}$ is the minimum on branch A and is the global floor; $P_{\star}$ is the minimum on branch B.

Both are roots of the irreducible sextic

$$8748x^6 - 49005x^4 + 27794x^2 + 18723 = 0,$$

whose real roots are exactly $\pm P_{\min}$ and $\pm P_{\star}$. Equivalently $P_{\min}^2 \approx 1.133599$ and $P_{\star}^2 \approx 4.856977$ are the two positive roots of the irreducible cubic $8748q^3 - 49005q^2 + 27794q + 18723 = 0$, whose third root is negative. The two floors are Galois conjugates. The critical cosines are the roots of $196c^3 + 224\sqrt{7}c^2 + 14c - 128\sqrt{7} = 0$ with $c \approx 0.673984$ (branch B) and $c \approx -0.924389$ (branch A). As $\mu \rightarrow 1$ the two branch minima merge into the classical equal-strength floor $\sqrt{2}$.

Disclaimer. This is the classical Gotoda / Aref unequal-circulation family: closed forms for its floors, not a new interaction law. The formula uses the product from Gotoda Prop. 2.1, which matches direct Biot–Savart on both arcs. An earlier campaign note built a different quotient from Gotoda’s eq. (3.3) for B ; that quotient disagrees with direct Biot–Savart (for example 1.746 against 2.207 at $\theta = 0.8$) and has a spurious minimum ≈ 1.741 . Whether the discrepancy lies in eq. (3.3) as printed or in that specialization has not been re-checked against Gotoda’s text.

1.1.1. Proof sketch

1. **Critical points.** Setting $dP/d\theta = 0$ and writing $c = \cos \theta$ gives the cubic above (checked symbolically). The substitution $c = \sqrt{7}u$ makes it rational: $1372u^3 + 1568u^2 + 14u - 128 = 0$, with exact real roots $u \approx -1.0482, -0.3494, 0.2547$.
2. **One critical point per arc.** Exact real-root isolation over $\mathbb{Q}$ puts exactly one root in each arc’s cosine range: $u \in (-1/14, 1/\sqrt{7})$ for branch B and $u \in (-1/\sqrt{7}, -1/14)$ for branch A. The third root has $|c| > 1$.
3. **Blow-up at the ends.** The numerator is positive at every arc end ($6\sqrt{7} + 21$ at $\theta \rightarrow 0^+$, $63/2$ at $\cos \theta = -\sqrt{7}/14$, and $21 - 6\sqrt{7} > 0$ at $\theta \rightarrow \pi^+$) while the denominator tends to 0 with the sign that makes $P \rightarrow +\infty$. With one critical point, each branch has a unique global minimum.
4. **Minimal polynomial.** An independent resultant elimination of c and $\sqrt{7}$ gives exactly $(8748P^6 - 49005P^4 + 27794P^2 + 18723)^2$. The sextic is irreducible over $\mathbb{Q}$ and over $\mathbb{Q}(\sqrt{7})$, and the cubic in $q = P^2$ has positive discriminant (casus irreducibilis), so the trigonometric Cardano forms cannot be reduced to real radicals. The branches $k = 0$ and $k = 1$ of $\cos(1/3 \arccos(X) - 2\pi k/3)$ give $P_{\star}^2$ and $P_{\min}^2$.

1.2. $n = 5$ polygon-collapse floor

Specialize the classical two-ring polygon collapse family (Koiller et al. 1985 §11) at $n = 5$: five vortices of strength 2 on the unit regular pentagon and five of strength -1 at radius $\sqrt{2}$, relative rotation θ . With the 2π kernel the configuration collapses self-similarly for every $0 < \theta < \pi/5$, and

$$\omega_0 t_c = \frac{127\sqrt{2} - 24 \cos(5\theta)}{80 \sin(5\theta)} \geq \sqrt{31682}/80,$$

with equality exactly when $\cos(5\theta_*) = 12\sqrt{2}/127$. So

$$F_5 = \sqrt{31682}/80 = 2.224929774172659\dots,$$

the $n = 5$ case of

$$F_n = \frac{\sqrt{K_n^2 - (2n-1)}}{2n}, \quad K_n = (n-1) \sinh\left(\frac{n+2}{2} \operatorname{arcosh} \frac{n}{n-1}\right),$$

alongside $F_2 = 3\sqrt{5}/4$, $F_3 = \sqrt{29}/3$ and $F_4 = \sqrt{322}/9$. This is not a new dynamical family.

1.2.1. Proof sketch

1. The inner strength is $x_5 = e^{\operatorname{arcosh}(5/4)} = 2$, so $K_5 = 2(2^{7/2} - 2^{-7/2}) = 127\sqrt{2}/8$ exactly.
2. The product $\omega_0 t_c = (K_5 - 3 \cos(5\theta))/(10 \sin(5\theta))$ clears to the quotient above.
3. The elementary bound $(a - b \cos \alpha)/\sin \alpha \geq \sqrt{a^2 - b^2}$ on $0 < \alpha < \pi$, with $a = 127\sqrt{2}$ and $b = 24$, has equality at $\cos \alpha = b/a$ and gives $\sqrt{31682}/80$. Consistently, $F_5 = \frac{\sqrt{K_5^2 - 9}}{10}$.

2. Independent validation (version 2)

Script: `research/verify_floors_independent.py` (Python, mpmath and sympy, about 25 s). Results: `research/artifacts/verify-floors-independent-2026-09-23.json`. The check does not reuse Gotoda's rates, the repo's JavaScript verifiers or the formulas above. It computes each vortex velocity from the 2π Biot–Savart law. A configuration collapses self-similarly exactly when every velocity is $\kappa(z_j - z_c)$ for one complex κ about the centre of vorticity. Then $|z|^2$ shrinks linearly and $\omega_0 t_c = |\operatorname{Im} \kappa| / (-2 \operatorname{Re} \kappa)$.

Check (60-digit arithmetic unless stated)	Result
Every $L = 0$ triangle for $\Gamma = (1, 1/2, -1/3)$ (a circle of shapes, 4000 samples): self-similarity residual	$\leq 9 \times 10^{-61}$ relative; exactly half the circle collapses
Minima of $\omega_0 t_c$ over the collapsing part	two branches: 1.0647059762712043373549... (counter-clockwise $1 \rightarrow 2 \rightarrow 3$) and 2.2038550160361327939941... (clockwise)
Closed forms above versus those minima	agree to all 40 digits printed
$P(\theta)$ versus raw Biot–Savart at Gotoda's own positions, 399 points on each arc	$\leq 10^{-59}$ relative on both; every branch-A point collapses

Both branch minima as $\mu \rightarrow 1$ ($\mu = 0.9, 0.99, 0.999, 1$)	(1.3399, 1.4983), (1.4067, 1.4218), (1.41346, 1.41496), then both $\sqrt{2}$; μ and $1/\mu$ give identical values
Time integration of the three-vortex ODE (Taylor method, 30 digits) at the $P_{\min}$ configuration	size ² follows the self-similar line to $\leq 5 \times 10^{-31}$ relative up to $0.9t_c$
Quotient built from Gotoda (3.3)– B versus raw Biot–Savart	disagrees (2.269 vs 3.118, 1.746 vs 2.207, 2.350 vs 2.705 at $\theta = 0.4, 0.8, 1.2$)
Exact algebra (sympy): critical cubic, one root per arc, resultant, irreducibility over $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{7})$, positive discriminant	all confirmed
Two-ring family, $n = 2, \dots, 8$: self-similar for every θ ; minimum over θ versus the F_n formula	residual $\leq 10^{-60}$; minima match F_n to $\leq 2 \times 10^{-60}$
$n = 5$ curve versus $(127\sqrt{2} - 24 \cos 5\theta)/(80 \sin 5\theta)$; $F_2, \dots, F_5$ radicals	$\leq 10^{-58}$ relative; radicals match to $\leq 10^{-34}$
$K_5 = 127\sqrt{2}/8$, $F_5 = \sqrt{31682}/80$, $\cos 5\theta_* = 12\sqrt{2}/127$ (exact)	confirmed symbolically

The check covers the stated families only. It does not show that no other configuration class for these circulations collapses more slowly per turn.

3. Literature (floor-kill and priority)

Source	Access	Verdict
Koiller et al. (1985)	Full text read	DOES NOT KILL optimized product floors
Aref, Phys. Fluids 25 (1982)	Full text (purchased PDF, all pages re-read for v2)	DOES NOT KILL. Sets up the general two-ring ($2n$ -vortex) center-of-symmetry equations (strengths κ , λ), of which the F_n family is a special case, but analyzes only $\kappa = -\lambda$ (Havelock double alternate rings: relative equilibria and pair dissolution). No collapse, no $\omega_0 t_c$, no floor.
O’Neil, Physica D 236 (2007)	Full text (purchased PDF, re-read for v2)	DOES NOT KILL. Finiteness of relative-equilibrium and collapse configurations of three rings; cites two-ring collapse as known (Koiller et al. 1985). No $\omega_0 t_c$ minimization.
Outside-family pass (web, OA, arXiv needles; GENChase PR #128)	Open web	STILL OPEN for P_* and F_5

Gröbli (1877) §10; Aref, Phys. Fluids 22 (2010)	repo reads (earlier passes)	Give the rates and the product $\omega_0 t_c$ as a logarithmic-spiral pitch for general strengths. $P(\theta)$ is a specialization of these; no $\mu = 1/2$ minimization located.
Tavantzis & Ting (1988)	search metadata	One-parameter contracting family in side-length variables, where orientation is invisible; consistent with two oriented arcs. DOES NOT KILL.
Leoncini, Kuznetsov & Zaslavsky (2000)	search metadata	Detailed analysis for two identical vortices only; excludes $\Gamma = (1, 1/2, -1/3)$.
Krishnamurthy & Stremmer (2018)	earlier postprint read; search metadata	Orientation-aware geometric collapse theory; collapse time and path length, no minimized $\omega_0 t_c$ found. Not re-read this pass.
Novikov & Sedov (1979)	search metadata	Parallelogram collapse; the $n = 2$ member of the two-ring family is classical. Its minimum F_2 is not claimed as new.
Demina & Kudryashov (2014)	abstract only	Explicit double-ring configurations of two regular polygons with arbitrary circulations. AMBIGUOUS for F_n and F_5 until read in full ; no statement of the minimization seen.
Fingerprint pass v2 (decimals of P_* , $P_{\min}$, F_5 ; 31682; 245351, 5201; sextic coefficients)	search engine only	No hits. Full-text pages and OEIS could not be opened from the checking machine, so no OEIS check is claimed.

Verdicts (version 2). P_* : **STILL OPEN**. $P_{\min}$: **STILL OPEN**. F_5 : **STILL OPEN**, with one unread lead (Demina & Kudryashov 2014) that could settle the general F_n minimization.

Limits. Aref (1982) and O’Neil (2007) were re-read in full for version 2. The other version 2 rows rest on search-engine metadata and snippets plus earlier full reads recorded in the repo; full texts and OEIS could not be opened from the checking machine. Negative searches are not a novelty certificate. Priority remains unconfirmed. The collapse families and the product itself are classical: the contribution claimed is only the minimizations and their closed forms.

Copyrighted PDFs are not redistributed with this note.

4. What would still kill priority

Any prior publication of the closed or minimal-polynomial form of $P_{\min}$ or P_* , or their decimals as product floors, of the two-branch structure of the $\mu = 1/2$ floor, of $F_5 = \sqrt{31682}/80$ or an equivalent, of the same Gotoda $\mu = 1/2$ minimization, or of the same K_5 clearance into $\sqrt{a^2 - b^2}$, would kill or demote priority while leaving the algebraic checks intact.

5. Suggested citation

Closed forms for the two $\mu = 1/2$ three-vortex product floors $P_{\min} \approx 1.064706$ (global) and $P_* \approx 2.203855$ (Gotoda's $L = 0$ branch), and for the $n = 5$ polygon floor $F_5 = \sqrt{31682}/80$, were derived by Chase Hendrick (2026) in the SharpMeow/GENChase validation campaign. Status: proved candidates, priority unconfirmed. Classical families only.

6. Anti-claims

- No novelty claim beyond “closed form / explicit radical written down and checked in this campaign.”
- No personal named theorem.
- No `IDENTITIES.md` edit from this file.
- Literature verdicts are floor-kill only.

7. References

1. H. Aref, Point vortex motions with a center of symmetry, *Phys. Fluids* **25**(12), 2183–2187 (1982). DOI 10.1063/1.863710.
2. H. Aref, Self-similar motion of three point vortices, *Phys. Fluids* **22**, 057104 (2010). DOI 10.1063/1.3425649.
3. M. V. Demina and N. A. Kudryashov, Rotation, collapse, and scattering of point vortices, *Theor. Comput. Fluid Dyn.* **28**, 357–368 (2014). DOI 10.1007/s00162-014-0319-4. (Abstract only.)
4. T. Gotoda, Self-similar motions and related relative equilibria in the N-point vortex system, *J. Dyn. Diff. Equat.* **33**, 1759–1777 (2021). DOI 10.1007/s10884-020-09867-y; arXiv:2002.09624.
5. W. Gröbli, Specielle Probleme über die Bewegung geradliniger paralleler Wirbelfäden (1877); English translation: arXiv:2404.01305.
6. J. Koiller, S. Pinto de Carvalho, R. Rodrigues da Silva and L. C. Gonçalves de Oliveira, On Aref's vortex motions with a symmetry center, *Physica D* **16**, 27–61 (1985). DOI 10.1016/0167-2789(85)90084-3.
7. V. S. Krishnamurthy and M. A. Stremler, Finite-time collapse of three point vortices in the plane, *Regul. Chaotic Dyn.* **23**(5), 530–550 (2018). DOI 10.1134/S1560354718050040.
8. X. Leoncini, L. Kuznetsov and G. M. Zaslavsky, Motion of three vortices near collapse, *Phys. Fluids* **12**(8), 1911–1927 (2000); arXiv:physics/9908055.
9. E. A. Novikov and Yu. B. Sedov, Vortex collapse, *Sov. Phys. JETP* **50**(2), 297–301 (1979).
10. K. A. O'Neil, Relative equilibrium and collapse configurations of heterogeneous vortex triple rings, *Physica D* **236**(2), 123–130 (2007). DOI 10.1016/j.physd.2007.07.015.
11. J. Tavantzis and L. Ting, The dynamics of three vortices revisited, *Phys. Fluids* **31**(6), 1392–1409 (1988).

12. SharpMeow/GENChase (2026-09-23): research/verify_floors_independent.py, identities/sources/unequal-mu-half-Pstar-closed-2026-09-23.md, new-formula-candidate-2026-09-23.md, deeper-kill-Pstar-F5-2026-09-23.md, outside-family-prior-art-Pstar-F5-2026-09-23.md.